# DR. PRIYA N. RAGHUNATHAN

ASSOCIATE PROFESSOR OF COMPUTATIONAL BIOLOGY

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## RESEARCH INTERESTS

Statistical and machine-learning methods for single-cell and spatial transcriptomics; probabilistic models of gene regulatory networks; reproducible computational pipelines for clinical genomics.

## EDUCATION

**University of California, Berkeley** 2008 – 2013

Ph.D. in Computational Biology

Dissertation: Hierarchical Bayesian models of transcriptional noise. Advisor: Prof. M. Lindqvist.

**University of Michigan** 2006 – 2008

M.S. in Statistics

**Indian Institute of Technology Madras** 2002 – 2006

B.Tech. in Biotechnology

## ACADEMIC APPOINTMENTS

**Associate Professor (with tenure)** Jul 2021 – Present

Carnegie Mellon University, Computational Biology Department — Pittsburgh, PA

- Lead a group of 7 Ph.D. students, 2 postdocs and 3 research staff.
- Courtesy appointment in Machine Learning Department (2022 - present).

**Assistant Professor** Jul 2016 – Jun 2021

Carnegie Mellon University, Computational Biology Department — Pittsburgh, PA

- Established the lab; graduated 3 Ph.D. students.

**Postdoctoral Fellow** Sep 2013 – Jun 2016

Broad Institute of MIT and Harvard — Cambridge, MA

- Single-cell methods group; Jane Coffin Childs Fellow.

## JOURNAL ARTICLES

**Spatially resolved gene regulatory networks from sparse transcriptomic maps** 2025

Raghunathan P.N.*, Okoye T., Brandt L., Feld S. Nature Methods 22(4):511-523

**Calibrated uncertainty for cell-type deconvolution** 2024

Okoye T., Liang Y., Raghunathan P.N. Genome Biology 25:188

**scNoise: a hierarchical model of transcriptional bursting at single-cell resolution** 2023

Liang Y., Raghunathan P.N. Cell Systems 14(2):97-110

**Benchmarking spatial clustering methods across 42 tissue atlases** 2022

Brandt L., Herrera A., ..., Raghunathan P.N. Nature Communications 13:6120

**Reproducible clinical genomics pipelines with provenance tracking** 2021

Feld S., Raghunathan P.N. Bioinformatics 37(19):3311-3318

**Variational inference for allele-specific expression in tumors** 2020

Herrera A., Kim J., Raghunathan P.N. PLOS Computational Biology 16(8):e1008021

**Gene-gene covariance from droplet-based scRNA-seq** 2019
Raghunathan P.N., Lindqvist M. Annals of Applied Statistics 13(3):1601-1630

**Single-cell atlas of the developing human thymus** 2018
Thymus Atlas Consortium (incl. Raghunathan P.N.). Science 367(6480):eaay3224

**Noise-aware differential expression for low-input RNA-seq** 2017
Raghunathan P.N., Weiss D., Regev-Haddad K. Genome Research 27(9):1525-1536

**Hierarchical Bayesian models of transcriptional noise** 2015
Raghunathan P.N., Lindqvist M. Journal of the American Statistical Association 110(512):1406-1419

## CONFERENCE PROCEEDINGS

Okoye T., Raghunathan P.N. Optimal transport for spatial alignment of serial sections. ICML 2024, pp. 38112-38129., Liang Y., Okoye T., Raghunathan P.N. Contrastive pretraining for gene programs. RECOMB 2023, LNCS 13976:201-216., Brandt L., Raghunathan P.N. Graph neural networks for tissue niches. ISMB 2022 (Bioinformatics 38 Suppl 1)., Herrera A., Raghunathan P.N. Amortized inference for clonal deconvolution. NeurIPS 2020, pp. 11843-11855., Raghunathan P.N. Sparse factor models for single-cell multi-omics. ISMB 2017 (Outstanding Paper Award)., Raghunathan P.N., Weiss D. Bursting kinetics from snapshot data. RECOMB 2016, LNCS 9649:254-268.

## GRANTS & FUNDING

**R01 GM148812: Probabilistic models of spatial gene regulation** 2023 – 2028
National Institutes of Health (NIGMS). PI. $2,450,000

**CAREER: Uncertainty-aware machine learning for single-cell genomics** 2019 – 2025
National Science Foundation (DBI-1945xxx). PI. $812,000

**Chan Zuckerberg Initiative Single-Cell Biology Data Insights** 2022 – 2024
CZI. Co-PI (with S. Feld). $400,000

**Sloan Research Fellowship** 2018 – 2020
Alfred P. Sloan Foundation. PI. $75,000

**Pilot: Clinical-grade pipelines for pediatric tumor sequencing** 2017 – 2018
UPMC Hillman Cancer Center. Co-I. $150,000

## TEACHING

**02-710 Computational Genomics (graduate)** 2017 – Present
Carnegie Mellon University. Instructor, ~60 students/yr. FCE 4.7/5

**02-251 Great Ideas in Computational Biology (undergraduate)** 2019 – Present
Carnegie Mellon University. Instructor, ~120 students/yr

**02-750 Automation of Scientific Research (graduate seminar)** 2021 – 2023
Carnegie Mellon University. Co-instructor

**Workshop: Bayesian Methods for Single-Cell Data** 2018, 2022
Cold Spring Harbor Laboratory course. Lecturer

## INVITED TALKS

**Keynote: Uncertainty is a feature, not a bug, in spatial omics** 2025
Single Cell Genomics Conference, Stockholm

**Spatial gene regulation from sparse maps** 2024
Department Seminar, Stanford University

**Calibrated deconvolution for clinical samples** 2024
Broad Institute Models, Inference & Algorithms

**Probabilistic programming for genomics** 2023
Gordon Research Conference: Quantitative Genetics

**Transcriptional bursting at scale** 2022
EMBL Heidelberg Seminar Series

**Benchmarks that matter** 2022
Human Cell Atlas General Meeting

**Noise models for single-cell data** 2019
Joint Statistical Meetings, invited session

**Reproducibility in computational biology** 2018
NIH Data Science Symposium

## PROFESSIONAL SERVICE

Associate Editor, PLOS Computational Biology (2021 - present), Program Committee, ISMB (2019 - 2025), RECOMB (2021 - 2024), NeurIPS (2020 - 2023), NIH Study Section ad hoc reviewer, GCAT and MABS (2020 - 2024), Reviewer: Nature Methods, Genome Biology, Cell Systems, Bioinformatics, JASA, Chair, CMU Computational Biology Ph.D. Admissions (2022 - 2024), Co-organizer, Pittsburgh Single-Cell Symposium (2019, 2023)

## HONORS & SKILLS

**Technical Skills:** `R` `Python` `Stan` `PyTorch` `Nextflow` `Slurm/HPC` `scRNA-seq` `Spatial transcriptomics`

**Languages:** `English` `Tamil` `Hindi`

**Awards:** `NSF CAREER Award (2019)` `Sloan Research Fellowship in Computational & Evolutionary Molecular Biology (2018)` `CMU School of Computer Science Teaching Award (2020)` `ISMB Outstanding Paper Award (2017)` `Jane Coffin Childs Postdoctoral Fellowship (2013 – 2016)`